ACTIVE FILTERS & OSCILLATORS

Introduction :

→ Filter is a frequency selective device. It is a circuit which is used for selecting a particular band of frequency.

→ Filter can be designed which passive and as well as the active components.

→ Passive elements are resistance, capacitor & inductor.

→ If the circuit designed with RLC which is called Passive filter.

→ Active elements are transistors, OP-amps. If the circuit is designed with op-amp which is called as active filter.

→ In active filter resistors and capacitors are also be used to desire circuit.

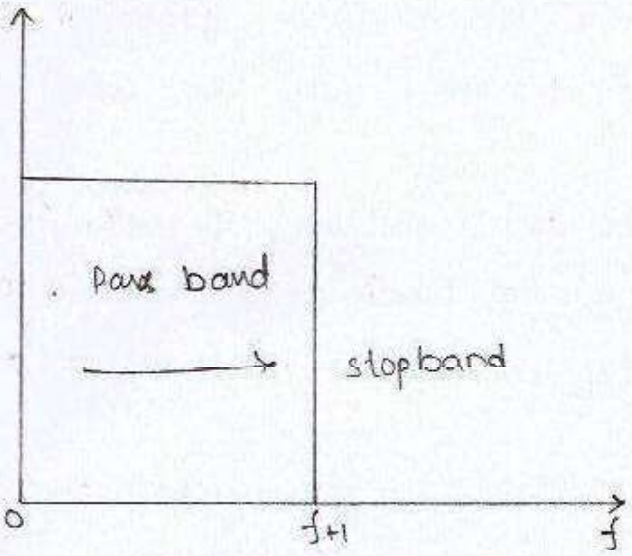
→ RC filters are used for low frequency oscillators. LC filters are used for high frequency oscillators.

Advantages of Active (or) Passive filters :

1. Gain and frequency adjustment flexibility.
2. No loading effect.
3. It is cheap (If is the cost is very low)
4. The most commonly used filters are
   (a) Low pass filter (LPF)
   (b) High pass filter (HPF)
   (c) Band pass filter (BPF)
   (d) Band Reject filter (BRF)
   (e) All Pass filter

Low Pass Filter :

In ideal low pass filter, the input signal frequency at the lower range of the band are allowed to pass and its completely stops after designed cutoff. frequency (f_H).



→ The low pass filter allows frequency for lower range of frequency upto f_H. and beyond f_H which is the cutoff frequencies, a higher cutoff frequency is totally stopped.

→ If we plot the gain Vs frequency of an ideal low pass filter we should get such type of characteristics.

→ The region or over the frequency band in which the signal is allow to pass i.e., called pass band.

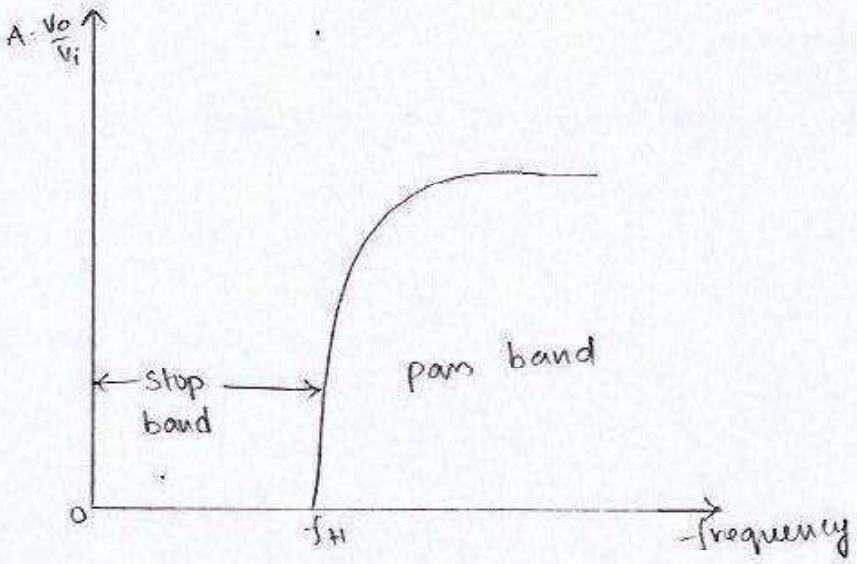
→ The frequencies beyond f_H is called stop band.

High Pass filter :

In high pass filter the higher band of frequency will

be allowed and lower band of frequencies will not be allowed.

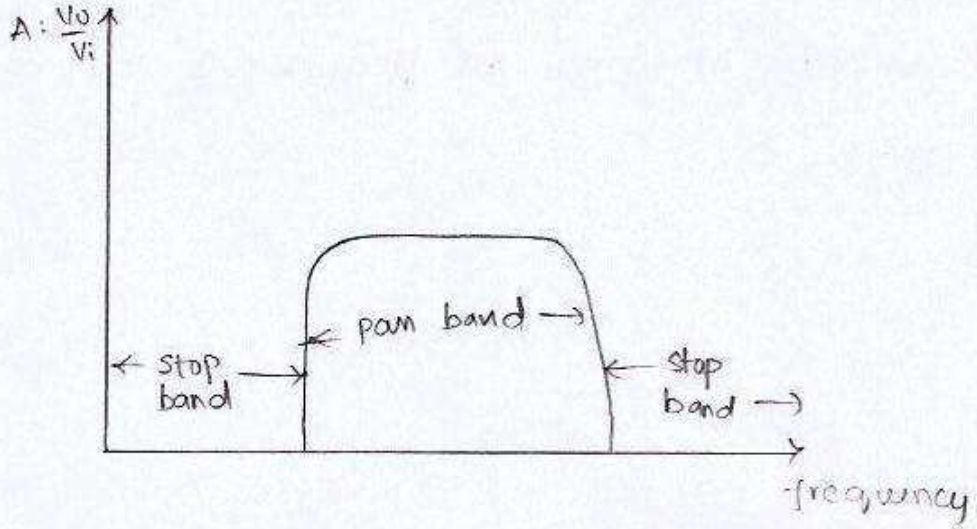
→ If we plot the gain vs frequencies.



Band Pass Filter :

The signal is passed with in a particular band which is called pass band.

→ Here between f_L and f_H band of frequencies is allowed but below f_L after f_H it is totally stoped.

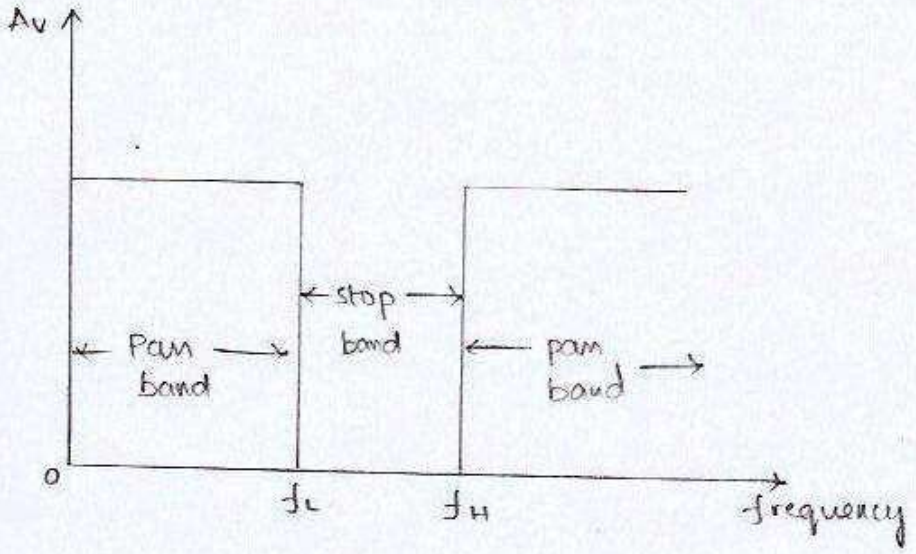


→ Basically band pass filter is a combination of high pass and low pass filters.

Band Reject Filter :

In band reject filter below the frequency band f_L is allow and above f_L and below f_H and frequency band is stopped.

→ Frequency band above f_H is allowed



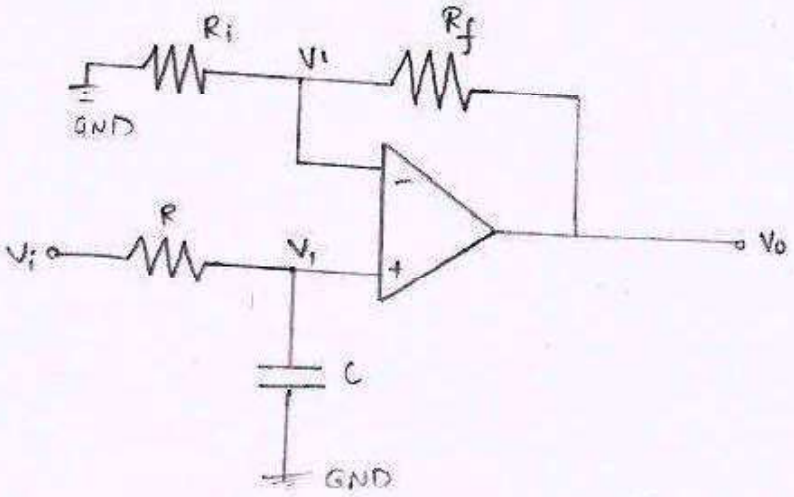
All pass Filter :

It will be allowing all frequencies to pass through it.

First Order LPF :

→ First order filter consists of a single RC network connected to the non-inverting input terminal of op-amp.

→ First order LPF shown in below figure.



→ Resistors R_f & R_i determines the gain of the filter.

→ The resistor R & capacitance C determines the cut off frequencies of the filter.

→ Since the op-amp is used in the non-inverting configuration, so the closed loop gain of the filter is given by

$$A_{VF} = 1 + \frac{R_f}{R_i}$$

Analysis :

→ We have to obtain the expression for at V_1 which is the voltage across the capacitor C.

→ The resistor R & capacitor C forms a voltage divider network across the i/p voltage Vi.

→ Therefore voltage V_1 at non-inverting terminal is given by according to potential divider theorem,

$$V_1 = \frac{V_i \cdot \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}}$$

$$= \frac{V_i \cdot \frac{1}{j\omega C}}{\frac{j\omega RC + 1}{j\omega C}}$$

$$\boxed{V_1 = \frac{V_i}{1 + j\omega RC}}$$

We know that, $A_{VF} = 1 + \frac{R_f}{R_i}$

$$\frac{V_o}{V_i} = 1 + \frac{R_f}{R_i}$$

$$V_o = \left(1 + \frac{R_f}{R_i}\right) V_i$$

$$= \left(1 + \frac{R_f}{R_i}\right) \frac{V_i}{1 + j\omega RC}$$

$$\frac{V_o}{V_i} = \left(1 + \frac{R_f}{R_i}\right) \frac{1}{1 + j\omega RC}$$

$$= A_{VF} \cdot \frac{1}{1 + j\omega RC}$$

But $\omega = 2\pi f$

$$\frac{V_o}{V_i} = A_{VF} \frac{1}{1 + j2\pi fRC}$$

$$= A_{VF} \frac{1}{1 + j\, f/\frac{1}{2\pi RC}}$$

$$= A_{VF} \frac{1}{1 + j\,(f/f_h)}$$

$\left[\because f_c = \frac{1}{2\pi RC}\right]$

$$\left|\frac{V_o}{V_i}\right| = \frac{A_{VF}}{\sqrt{1+(f/f_H)^2}}$$

The first order LPF can be verified from the above eqn as under 3 conditions.

1. At very low frequencies i.e., $f << f_h$

$$\left|\frac{V_o}{V_i}\right| \simeq A_{VF}$$

Thus at very low frequencies, the filter gain is constant.

2. At $f = f_h$

$$\left|\frac{V_o}{V_i}\right| = \frac{A_{VF}}{\sqrt{1+(1)^2}}$$

$$= \frac{A_{VF}}{\sqrt{2}}$$

$$\left|\frac{V_o}{V_i}\right| = 0.707\, A_{VF}$$

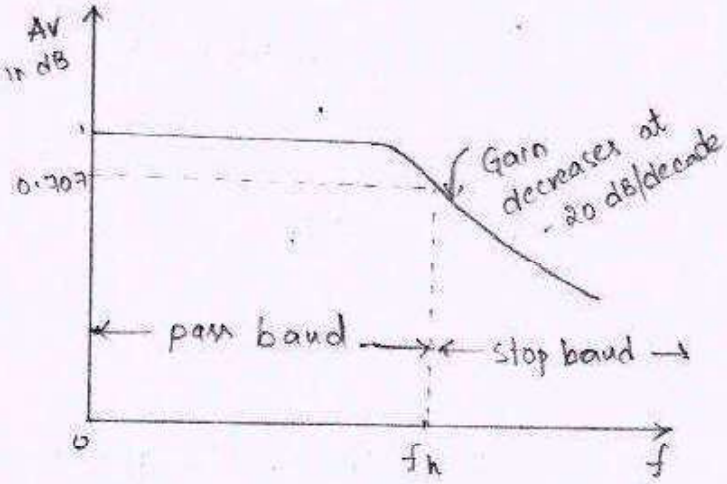
Thus at $f = f_h$, the filter gain reduces by 3 dB, as the dB value of 0.707.

3. At very high frequencies i.e., $f >> f_h$.

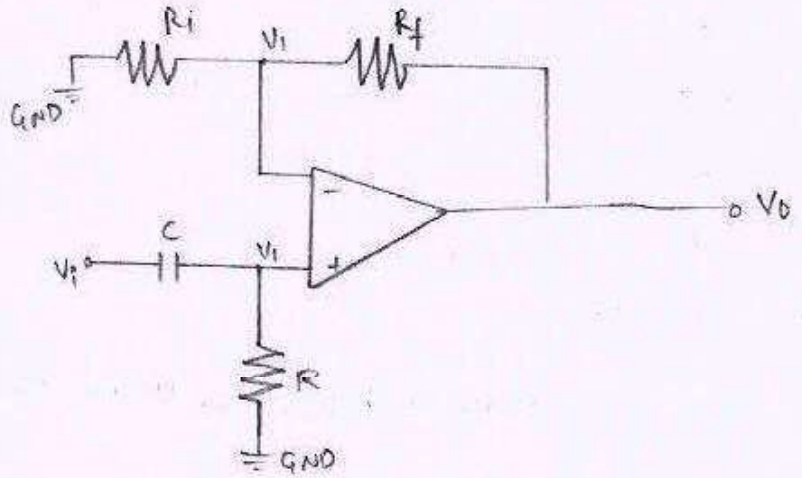
$$\left|\frac{V_o}{V_i}\right| < A_{VF}$$

Thus the filter gain will keep decreasing with increasing frequency. This decreases takes place at a constant rate of -20 dB/decade. Here "decade" means 10 times & the gain is expressed in dB. & -ve sign indicates that gain is decreasing.

→ Hence, frequency response of first order LPF is shown in below fig.



First Order HPF :



Analysis :

→ We obtain voltage across resistor R. i.e, V_1.

→ According to potential divider rule,

$$V_1 = V_i \cdot \frac{R}{R + \frac{1}{j\omega C}}$$

$$V_1 = V_i \frac{j\omega RC}{1 + j\omega RC}$$

W.K.T $\quad A_{VF} = 1 + \frac{R_f}{R_i}$

$$\frac{V_o}{V_i} = 1 + \frac{R_f}{R_i}$$

$$V_o = \left(1 + \frac{R_f}{R_i}\right) V_i$$

$$= \left(1 + \frac{R_f}{R_i}\right) \times V_i \frac{j\omega RC}{1 + j\omega RC}$$

$$\frac{V_o}{V_i} = \left(1 + \frac{R_f}{R_i}\right) \times \frac{j\omega RC}{1 + j\omega RC}$$

But $\omega = 2\pi f$

$$= \left(1 + \frac{R_f}{R_i}\right) \times \frac{j2\pi fRC}{1 + j2\pi fRC}$$

$$= \left(1 + \frac{R_f}{R_i}\right) \times \frac{j\, f/1/2\pi RC}{1 + j\, f/1/2\pi RC}$$

$$\frac{V_o}{V_i} = A_{VF} \times \frac{j\, f/f_L}{1 + j\, f/f_L}$$

$$\left|\frac{V_o}{V_i}\right| = \frac{A_{VF} \times (f/f_L)}{\sqrt{1 + (f/f_L)^2}}$$

$$= \frac{A_{VF}}{\sqrt{\frac{1 + (f/f_L)^2}{(f/f_L)^2}}}$$

$$\boxed{\left|\frac{V_o}{V_i}\right| = \frac{A_{VF}}{\sqrt{1 + (f/f_L)^2}}}$$

→ The first order HPF can be verified from the above eq" under 3 conditions.

1. At very low frequencies i.e $f << f_L$

$$\left|\frac{V_o}{V_i}\right| < A_{VF}$$

Thus at very low frequencies, gain is Low, with increase in frequency, gain also increases.

2. ~~At $f = f_H$~~

2. At $f = f_L$

$$\left|\frac{V_o}{V_i}\right| = \frac{A_{VF}}{\sqrt{2}}$$

$$= 0.707 A_{VF}$$

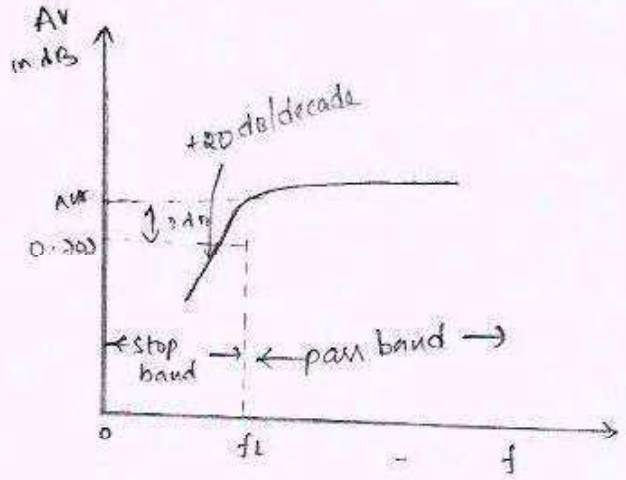
Thus at $f = f_L$ the filter gain increases by 3 dB.

3. At very high frequencies i.e., $f >> f_L$.

$$\left|\frac{V_o}{V_i}\right| \cong A_{VF}$$

→ Thus at very high frequencies the filter gain is constant.

→ Hence the frequency response of first order HPF shown in below figure.



Band Pass Filter:

→ A band pass filter has a pass band b/w two cut-off frequencies f_H & f_L. i.e., it is passes the frequencies f_H & f_L. Outside of this pass band will stopped the frequencies.

There are two types of band pass filter which are classified as per the quality factor.

(a) Wide band pass filter

(b) Narrow band

→ If the quality factor is less than 10 ($Q < 10$) then the filter is called as wide band pass filter.

→ If the quality factor is greater than 10 ($Q > 10$) then the filter is called as Narrow band pass filter.

Quality factor:

→ Quality factor Q is the measure of selectivity of filter. The value of Q is given by

$$Q = \frac{f_c}{f_H - f_L}$$

where, $f_H - f_L$ = Band width, so

$$Q = \frac{f_c}{B.W}$$

where $f_c = \sqrt{f_H f_L}$

f_H = upper cut-off frequency

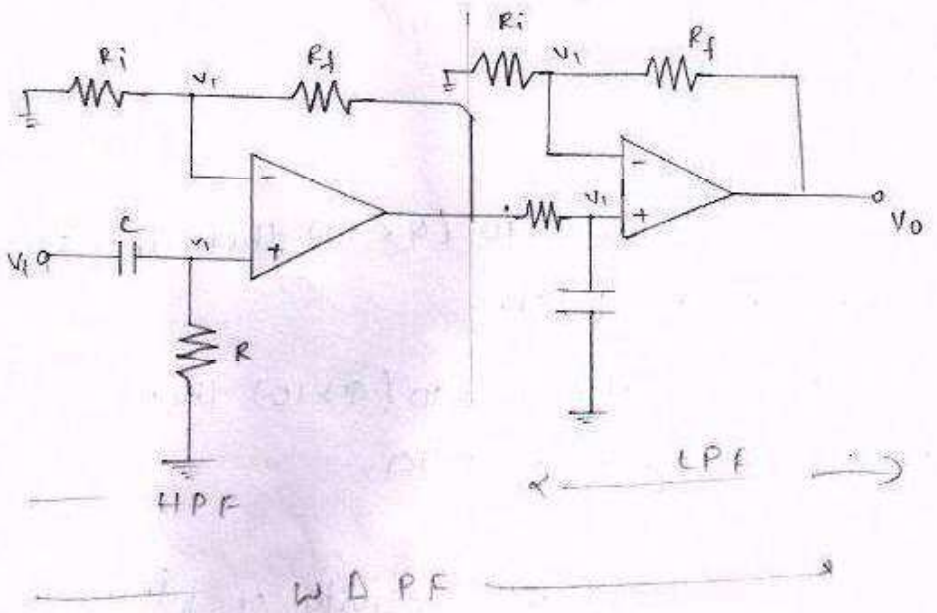
f_L = lower cut-off frequency

(a) Wide Band pass filter:

→ A wide band pass filter is implemented by cascading a first order HPF & first order LPF.

→ It has more band width b/w the two frequencies.

→ The circuit diagram of wide band pass filter shown in below figure.



→ Let us consider individual voltage gains of HPF & LPF as AVF_1 & AVF_2.

→ The overall voltage gain of the wide band pass filter is AV_F.

→ Then AV_F is the product of individual voltage gains.

i.e., $$AV_F = AVF_1 \times AVF_2$$

→ The magnitude voltage gain of first order HPF is given by

$$\left|\frac{V_o}{V_i}\right| = \frac{AVF_1}{\sqrt{1+(f_L/f)^2}}$$

→ The magnitude voltage gain of first order LPF is given by

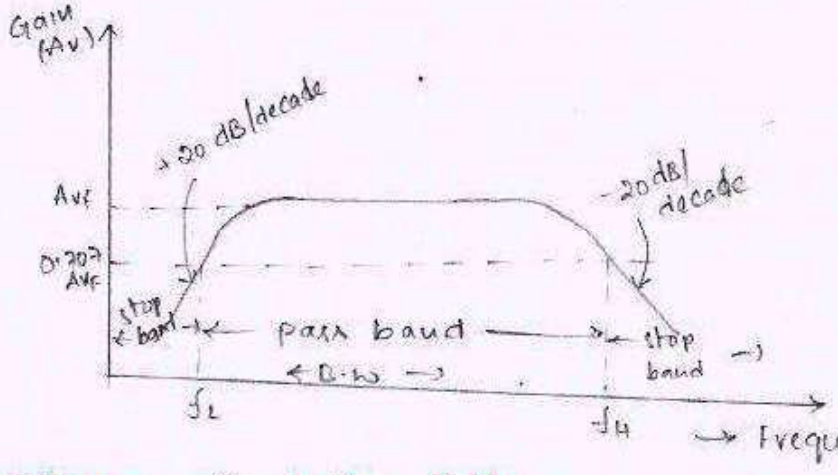
$$\left|\frac{V_o}{V_i}\right| = \frac{AVF_2}{\sqrt{1+(f/f_H)^2}}$$

→ Now, the magnitude voltage gain of wide band pass filter is the product of first order HPF & LPF.

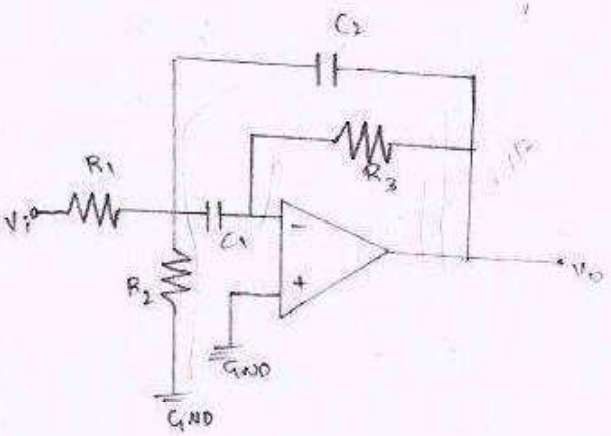
$$\left|\frac{V_o}{V_i}\right| = \frac{AVF}{\left[\sqrt{1+(f_L/f)^2}\right]\left[\sqrt{1+(f/f_H)^2}\right]}$$

where, $A_{VF} = A_{VF_1} \times A_{VF_2}$

→ The frequency response of wide band pass filter is shown in below fig.

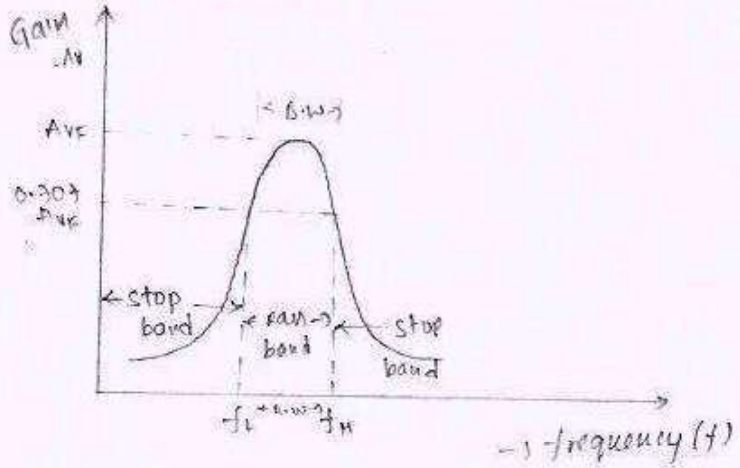


(b) Narrow Band Pass Filter :



→ The narrow band pass filter shown in below fig.

→ The narrow band pass filter is a band pass filter with a small band width shown in below fig.



→ The frequency response of narrow band pass filter is sharper than that of a wide band pass filter.

→ The quality factor of this filter is high than that of wide band pass filter.

→ The main features of this filter is

(a) It has only one op-amp

(b) It has two fb paths. Hence it is called as multipath fb filter.

(c) Here the op-amp is inverting mode.

(d) Here the Band width is small compared to WBPF.

Band Reject Filter :

→ Band reject filter is a filter in which stop the frequencies in stop band & passes all the frequencies outside the stop band.

→ It is also called as Band stop filter (or) Band elimination filter.

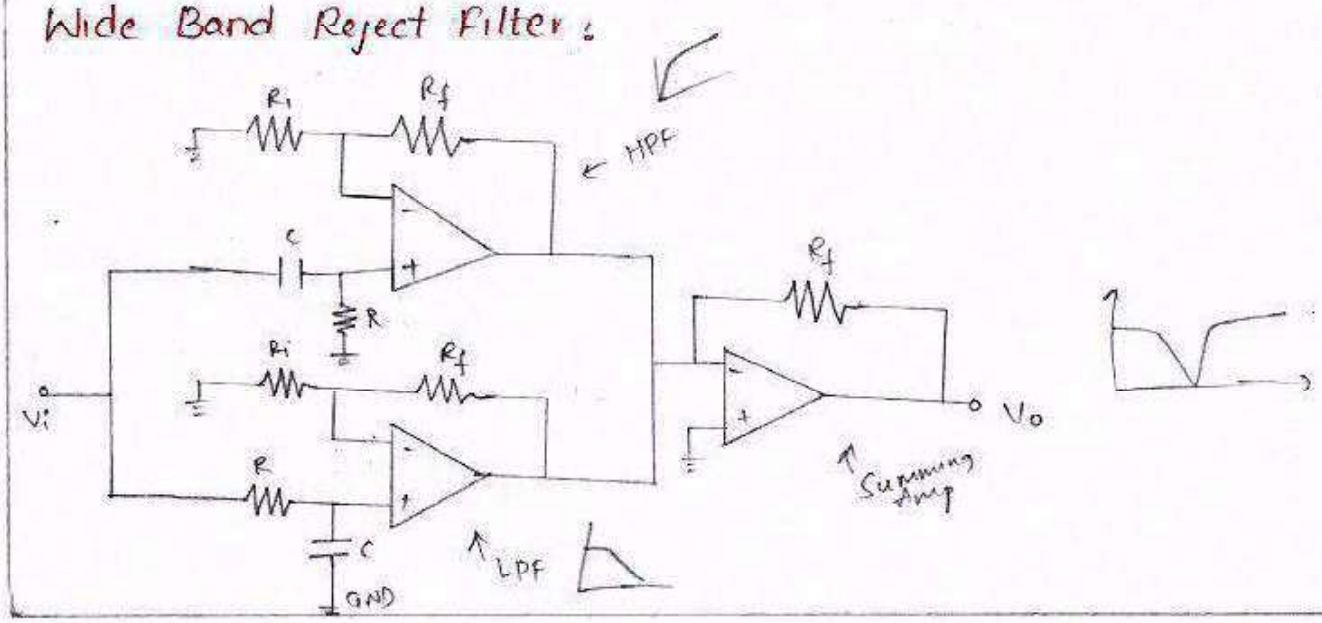
→ The band reject filter operation is exactly opposite to the band pass filter.

→ Band reject filter are classified into 2 types.

(a) Wide band reject filter

(b) Narrow band reject filter

Wide Band Reject Filter :



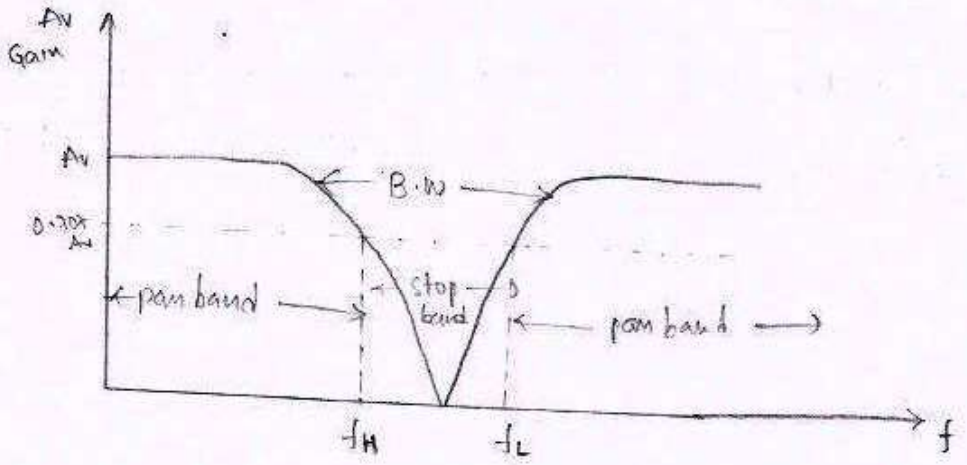
→ Like WBPF, this filter also consists of a first order HPF & LPF sections. Additionally it consists of a summing amplifier.

→ Here stops the all frequencies b/w f_H & f_L.

→ The lower cut-off frequency f_L must be greater than the higher cut-off frequency f_H. It is shown in frequency response of the filter. shown in below fig.

→ The pass band gain of both high pass & low pass sections must be equal.

→ Hence the frequency response of the filter is shown in below fig.



Narrow Band Reject Filter :

→ Narrow band reject filter is also called as the notch filter. The quality factor of such filters is higher than that of the WBRF.

→ Narrow band reject filter have a very very sharp frequency response characteristics.

→ These are used for rejecting a single frequency such as 50 Hz power line frequency hum.

→ Notch filter is the Twin-T n/w shown in below fig.

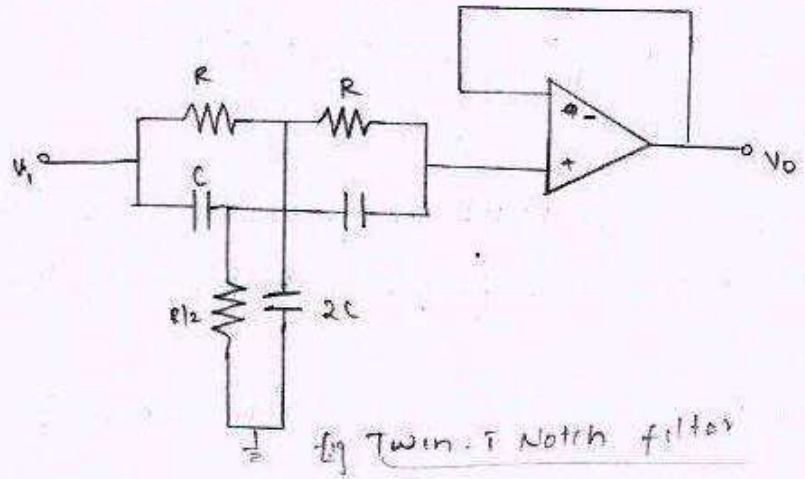
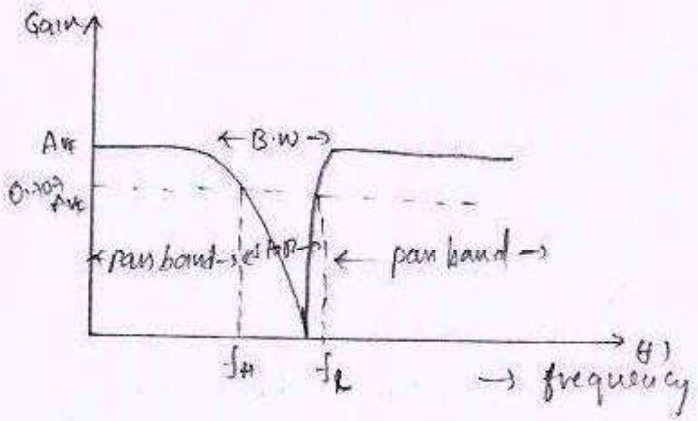


fig Twin-T Notch filter

→ The ckt consists of two T-networks. One consist of 2 resistors & 1 capacitor while other consist of 2 capacitor & 1 resistor.

→ To design a Notch filter to reject the particular frequency.

→ The frequency responce of the Notch filter shown in below fig.



Applications :-

1. Communication circuits
2. Biomedical instruments in order to eliminate the particular frequency.

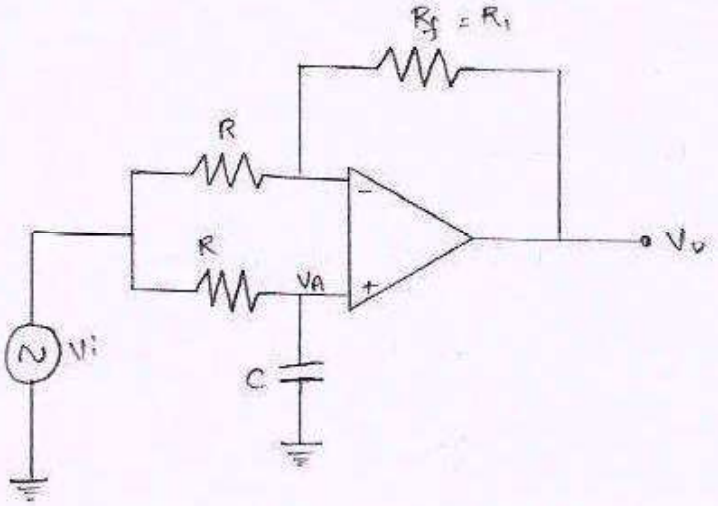
All Pass Filter :

→ APF passes all frequency components of the i/p signal without any attenuation but it provides 90° phase shift for different frequencies of the i/p signal.

→ Its magnitude of the gain is 1 (unity) for all the frequencies.

→ When signals are transmitted over the transmission lines there is change in their phase.

→ Hence all pass filters are also called as delay equalizer because circuit takes some delay to o/p signal.



→ Let us use super position principle to obtain the o/p V_o.

→ Assume i/p to the non-inverting terminal is zero. so CKt act as an inverting amplifier.

→ Therefore, voltage gain of inverting terminal

$$A_{VF} = \frac{-R_f}{R_1}$$

$$\frac{V_{o1}}{V_i} = \frac{-R_f}{R_1}$$

$$V_{o1} = \frac{-R_f}{R_1} \times V_i$$

as $R_f = R_1$

$$V_{o1} = -V_i \rightarrow (1)$$

→ similarly, let us assume i/p to the inverting terminal is '0'. So the circuit acts as an non-inverting amplifier.

$$\therefore A_{VF_2} = 1 + \frac{R_f}{R_1}$$

$$\frac{V_{o2}}{V_A} = 1 + \frac{R_f}{R_1}$$

$$V_{o2} = \left[1 + \frac{R_f}{R_1}\right] V_A$$

as $R_f = R_1$

$$V_{o2} = 2V_A \rightarrow (2)$$

→ The o/p voltage of the circuit is given by,

$$V_o = V_{o1} + V_{o2} \rightarrow (3)$$

Sub. (1) & (2) in eq(3), we get

$$\boxed{V_o = -V_i + 2V_A} \rightarrow (4)$$

→ By potential divider rule, the voltage V_a can be obtained as

$$V_A = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \times V_i$$

$$= \frac{\frac{1}{j2\pi fC}}{\frac{j2\pi fC + 1}{j2\pi fC}} \times V_i$$

$$V_A = \frac{1}{1+j2\pi fRC} \times V_i \quad \longrightarrow (5)$$

sub. eq(5) in eq(4)

$$V_o = -V_i + 2V_i\left[\frac{1}{1+j2\pi fRC}\right]$$

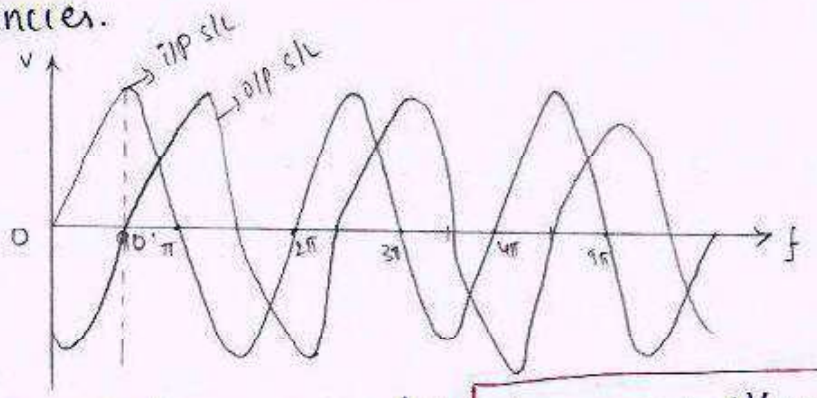
$$= V_i\left[-1 + \frac{2}{1+j2\pi fRC}\right]$$

$$\frac{V_o}{V_i} = \frac{1-j2\pi fRC}{1+j2\pi fRC}$$

$$\left|\frac{V_o}{V_i}\right| = \frac{\sqrt{1+(2\pi fRC)^2}}{\sqrt{1+(2\pi fRC)^2}}$$

$$\boxed{\left|\frac{V_o}{V_i}\right| = 1}$$

→ The above expression says that the magnitude voltage gain is always 1. for all pass filter and it can pass the range of frequencies.



phase angle is given by $\boxed{\phi = -2\tan^{-1}(2\pi RC)}$

→ This is the phase angle in degrees which indicates there is a phase shift of angle ϕ° b/w i/p & o/p signal.

Bark Hausen Criterion :

It states that

1. The total phase shift around a loop should be 0° or 360°.
2. The magnitude of the product of open loop gain of the op-amp and feedback factor (β) is unity.

i.e., $|A\beta| = 1$

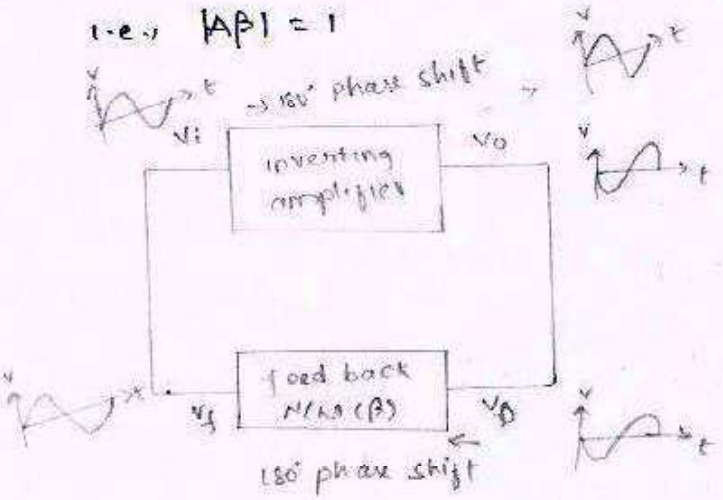


fig: Block Diagram of oscillator circuit

→ The flb n/w i/p is V_o, then the feedback n/w produces 180° phase shift.

→ This feedback signal is given to the i/p of the inverting amplifier then again 180° phase shyt is provided by the inverting amplifier. so total phase shift around a loop is 0° (or) 360°.

→ Let input voltage of the feedback n/w is V_o i.e o/p voltage of the inverting amplifier. It is given by

$$V_o = A \cdot V_i \qquad \left[\because A = \frac{V_o}{V_i}\right]$$

→ The feedback n/w β provides 180° phase shift i.e., given by

$$V_f = -\beta V_o \qquad \beta = V_f / V_o$$

$$V_f = \beta V_o$$

where, -ve sign indicates that 180° phase shift provided by the flb to network.

→ For the oscillator, V_f must acts as an i/p voltage V_i. So

$$V_i = -\beta V_o$$

we know that, $V_o = AV_i$

$$V_i = -\beta A V_i$$

$$-\beta A = 1$$

Let us take magnitude on b.s

$$\therefore |A\beta| = 1$$

→ The above condition is called as Bark Hausen criterion.

→ The phase of V_f must be as i/p voltage V_i i.e., feedback n/w introduces 180° phase shift. In addition to 180° phase shift introduced by the amplifier. so total phase shift around a loop is 360°. In this condition feedback voltage V_f drives the circuit without external i/p. so the circuit acts as an oscillator.

→ Similarly the magnitude of product of open loop gain & feedback factor is unity. It is also called as Bark Hausen Criterion.

→ The above 2 conditions are required to satisfy the circuit works as an oscillator producing sustained oscillations of constant frequency & amplitude.

→ Let us see the effect of magnitude of product of gain & flb factor $[|A\beta|]$ on the nature of the oscillations.

(i) $|A\beta| > 1$:

When the total phase shift around a loop is 0° or 360° and $|A\beta| > 1$ then the oscillations are growing type i.e., the amplitude of the oscillations goes on increasing. It is shown in fig.

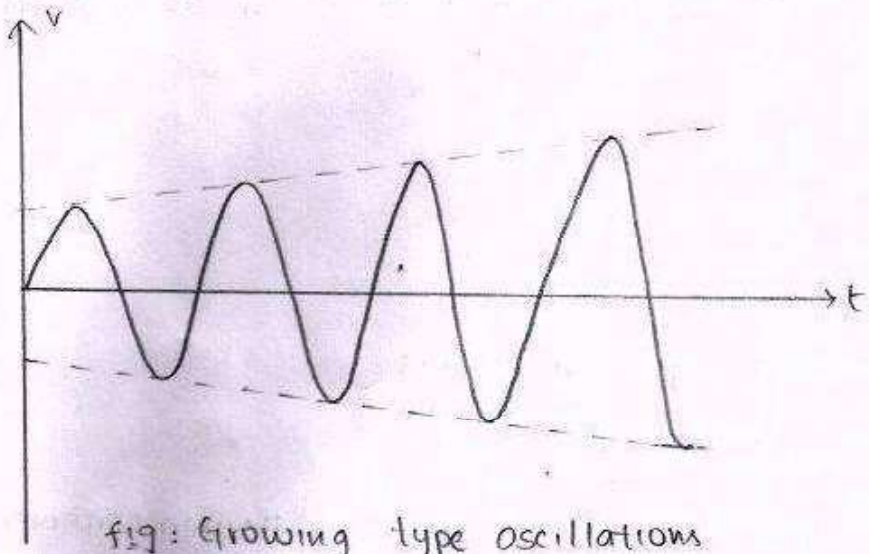


fig: Growing type oscillations

(ii) $|A\beta| = 1$:

When the total phase shift around a loop is 0° or 360° & $|A\beta| = 1$, then the oscillations are with constant frequency & amplitude, it is called as sustained oscillations. It is shown in below fig.

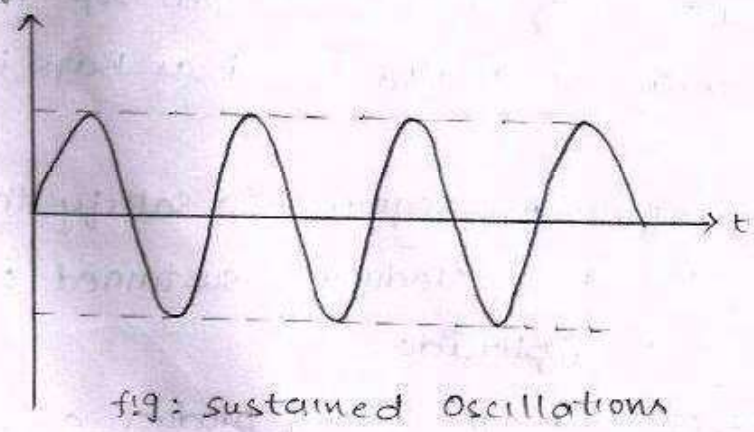


fig: sustained Oscillations

(iii) $|A\beta| < 1$:

When the total phase shift around a loop is 0° or) 360° & $|A\beta| < 1$ then the oscillations are decaying type i.e., the amplitude of the oscillations decreases exponentially. It is shown in below fig.

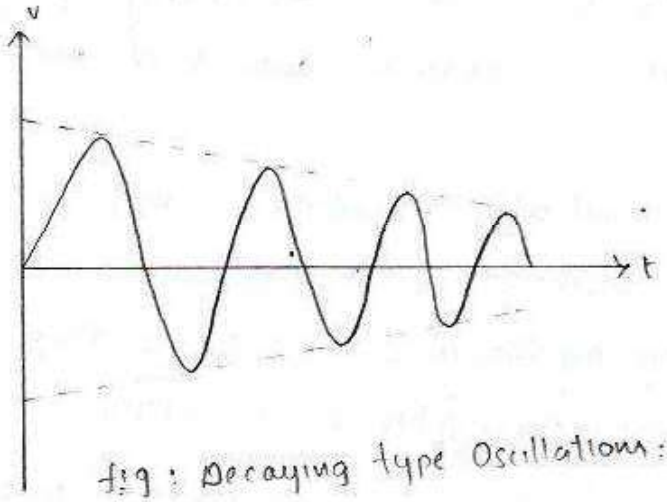


fig : Decaying type Oscillations.

Classification of Oscillations :

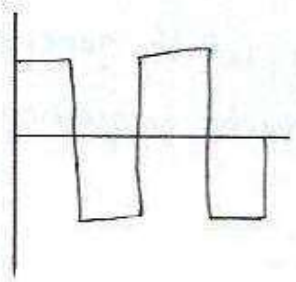
Oscillators are classified into different ways

(1) According to waveform
(2) According to frequency range
(3) According to RLC components.

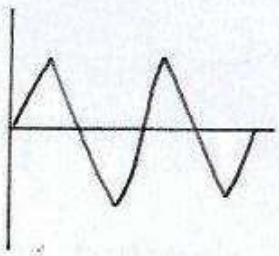
(1) **According to waveform :** According to waveform the oscillator can be classified into 2 types.

(a) **Sinusoidal oscillator** : A sinusoidal oscillator generates the sinusoidal signals only.

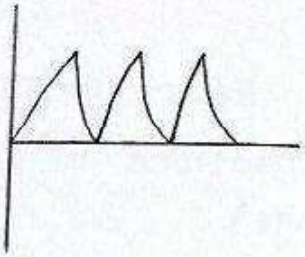
(b) **Non-sinusoidal oscillator** : It generates the non-sinusoidal signals like square, triangular & sawtooth.



Square waveform



Triangular waveform



Sawtooth waveform

(2) According to frequency range : According to frequency range the oscillator can also be classified into different types.

(a) Audio frequency oscillator :- 20 Hz - 20 KHz

(b) Radio frequency oscillator :- > 20 KHz

(c) Very high frequency oscillator :- 30 MHz - 300 MHz

(d) Micro wave frequency oscillator :- > 300 MHz

(3) According to RLC components : According to RLC components the oscillator can be classified into different ways.

(a) RC oscillators : The oscillator using the components R & C are called as RC oscillators. It can be used to generate the low frequency signals in audio range.

(b) LC oscillators : The oscillator using the components L & C are called as LC oscillators. It can be used to generate the high frequency signals.

(c) Crystal Oscillators : The oscillator using the crystal is called as crystal oscillator. It can be used to generate the very high frequency signals.

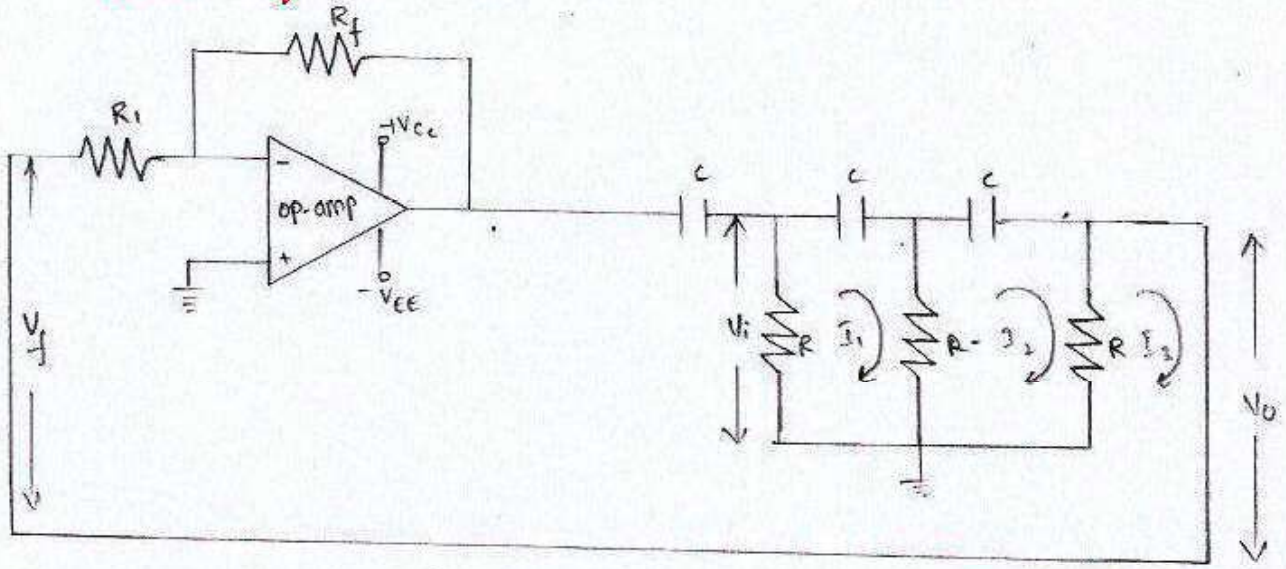
RC Oscillator :

RC oscillator is a one type of oscillator which generates oscillations in the low frequency range (audio frequency range).

→ RC oscillators can be classified into 2 types.

1. RC phase shift oscillator
2. Wein bridge oscillator

RC Phase Shift Oscillator :



→ It consists of CE amplifier followed by a 3 sections of RC phase shift network. In the RC phase shift n/w RC oscillator can be used as a feedback path.

→ In oscillator, amplifier produces 180° phase shift & feedback must introduced 180° phase shift to obtain a total phase shift around a loop is 360°.

→ One RC network produces phase shift of $\phi = 60°$.

→ Here 3 RC networks are available to produce phase shift is 180° (60° + 60° + 60°). The feedback network is also called as ladder network.

→ In this network all the resistances & capacitances values are same. so that for a particular frequency each section of RC produces a phase shift is 60°. The o/p of RC phase shift n/w is connected to the input of CE amplifier through RC feedback network.

→ To make 3 RC sections are identical (similar), R_3 should be choosen as

$$R_i' = R_i + R_3$$

$$R_i = R_i' - R_3$$

$$R_3 = R_i' - R_i$$

→ The phase shift ϕ produced by the RC section is

$$\phi = \tan^{-1}\left[\frac{1}{\omega RC}\right]$$

$$= \tan^{-1}\left[\frac{1}{2\pi f RC}\right]$$

Let us assume $R = 1K\Omega$, $C = 0.1\mu F$, $F = 1KHz$

$$\phi = \tan^{-1}\left[\frac{1}{2\pi \times 1\times10^{3}\times 1\times10^{3}\times 0.1\times10^{-6}}\right]$$

$$\phi = 57.85 \simeq 60^{\circ}$$

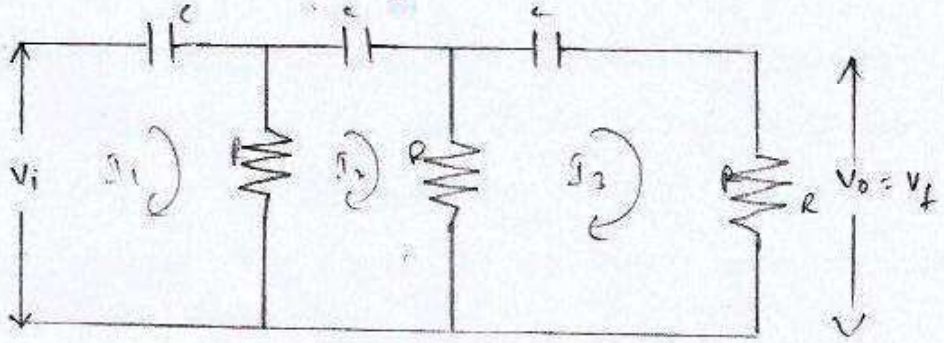
→ If all the resistors & capacitors are same in 3 sections then each section can produce a phase shift of 60°. So the ladder network produce 180° phase shift in between o/p & i/p voltages.

→ The total phase shift from the base of the op-amp around the circuit will be exactly 360°. so there by satisfying Bark-Hausen Criterion for Oscillations.

→ The frequency generated by the RC phase shift oscillator is

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

Derivation for frequency of Oscillations :



Apply KVL for 1st, 2nd and 3rd loop

loop (1) :- $V_i = I_1\left(\frac{1}{j\omega C}\right) + R(I_1 - I_2)$

$$= I_1\left[\frac{1}{j\omega C} + R\right] - R I_2 \rightarrow (1)$$

loop(2): $0 = I_2\left(\frac{1}{j\omega C}\right) + R(I_2 - I_1) + R(I_2 - I_3)$

$= I_2\left(\frac{1}{j\omega C} + 2R\right) - RI_1 - RI_3$

$0 = -RI_1 + I_2\left(\frac{1}{j\omega C} + 2R\right) - RI_3 \rightarrow (2)$

loop(3): $V_o = I_3\left(\frac{1}{j\omega C}\right) + R(I_3 - I_2) + RI_3$

$= I_3\left(\frac{1}{j\omega C} + 2R\right) - RI_2$

$V_o = -RI_2 + I_3\left(\frac{1}{j\omega C} + 2R\right) \rightarrow (3)$

By solving the 3 eqns, we get

$$I_3 = \frac{V_i}{R}\left[\frac{1}{(1-5\alpha^2) + j\alpha(\alpha^2-6)}\right]$$

$$\alpha = \frac{1}{\omega RC} \quad (or) \quad \alpha = \frac{1}{2\pi f RC}$$

W.K.T ; $V_o = I_3 R$

$$V_o = V_i\left[\frac{1}{(1-5\alpha^2) + j\alpha(\alpha^2-6)}\right]$$

for determining the frequency of an oscillator the imaginary part must be equal to zero

i.e., $\alpha(\alpha^2-6) = 0$

$\alpha^2 - 6 = 0$

$\alpha^2 = 6$

$\alpha = \pm\sqrt{6}$

W.K.T, $\alpha = \frac{1}{\omega RC}$

$\frac{1}{\omega RC} = \sqrt{6}$

$\frac{1}{2\pi f RC} = \sqrt{6}$

$f = \frac{1}{2\pi f RC\sqrt{6}}$

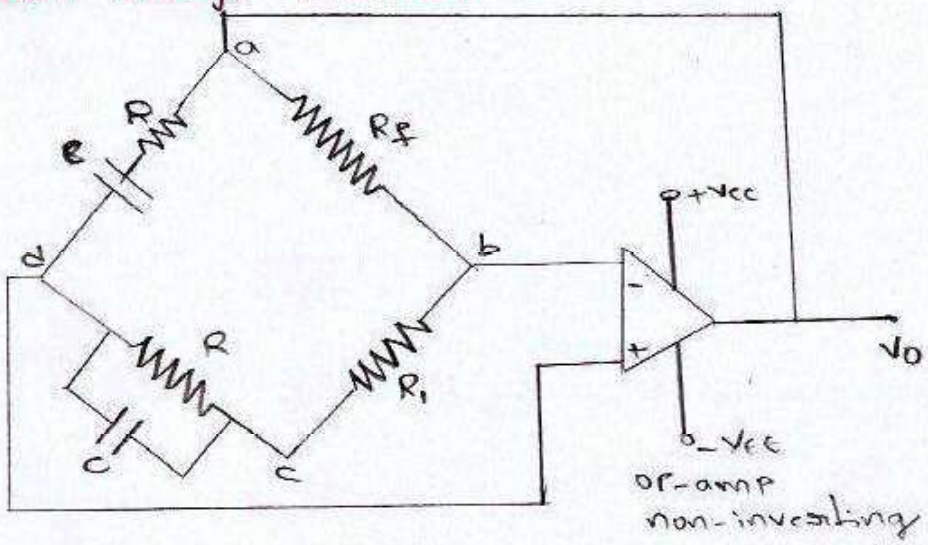
Advantages :

1. The circuit is simple to design.
2. It is suitable to produce o/p waveform for audio frequencies only.
3. It produces sinusoidal o/p voltages.
4. It is able to generate the low frequency signals.

Disadvantages :

1. By changing the values of R & C, the frequency of oscillator can be changed.
2. It is unable to generate the high frequency signals.

(2) Wein Bridge Oscillator :



→ Generally in an oscillator the amplifier stages introduces 180° phase shift and feedback network introduces another 180° phase shift. To obtain a phase shift of 360° around a loop.

→ But wein-bridge oscillator consist of non-inverting amplifier & hence does not provide any phase shift during amplifier stage.

→ As the total phase shift required is 360° in wein bridge, because no phase shift is necessary.

→ The o/p of amplifier is connected b/w terminals a & c which is i/p.

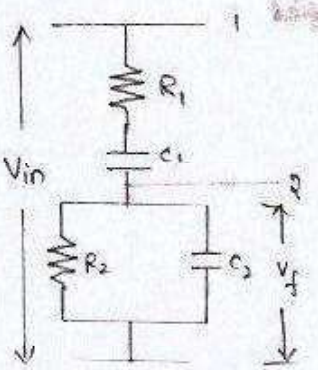
→ While the amplifier i/p is connected b/w terminals b & d which is o/p.

→ The bridge n/w consists of 2 arms namely R_1C_1 in series & R_2C_2 in parallel. These arms are called frequency sensitive arms which decides the frequency.

→ The frequency of wein bridge oscillator

$$f = \frac{1}{2\pi RC}$$

→ This is also called as lead lag n/w because at low frequency signals the voltage leads current.



$$Z_1 = R_1 + \frac{1}{j\omega C_1}$$

$$= \frac{j\omega R_1 C_1 + 1}{j\omega C_1}$$

$$Z_2 = \frac{R_2 \times 1/j\omega C_2}{R_2 + \frac{1}{j\omega C_2}}$$

$$= \frac{R_2/j\omega C_2}{\frac{j\omega C_2 R_2 + 1}{j\omega C_2}}$$

$$= \frac{R_2}{1 + j\omega C_2 R_2}$$

$$Z_1 = \frac{1 + sR_1C_1}{sC_1} \quad ; \quad Z_2 = \frac{R_2}{1 + sC_2R_2}$$

From fig. $I = \frac{V_{in}}{Z_1 + Z_2}$ i.e., $V_f = I \times Z_2$

$$V_f = \frac{V_{in} \times Z_2}{Z_1 + Z_2}$$

We know that, $\beta = \frac{V_f}{V_{in}}$

$$= \frac{\frac{V_{in} \times Z_2}{Z_1 + Z_2}}{V_{in}}$$

$$= \frac{Z_2}{Z_1 + Z_2}$$

$$\beta = \frac{SC_1R_2}{1 + SC_1R_1(1 + SC_2R_2) + SC_1R_2}$$

$$= \frac{SC_1R_2}{1 + SC_2R_2 + SC_1R_1 + S^2C_1R_1C_2R_2 + SC_1R_2}$$

$$= \frac{SC_1R_2}{1 + S(C_2R_2 + C_1R_1 + C_1R_2) + S^2C_1C_2R_1R_2}$$

$$= \frac{j\omega C_1R_2}{1 + j\omega(C_2R_2 + C_1R_1 + C_1R_2) - \omega^2C_1C_2R_1R_2}$$

$$= \frac{R_2 / SC_2R_2 + 1}{\frac{(1 + SC_1R_1)(1 + SR_2C_2) + R_2SC_1}{SC_1(1 + SC_2R_2)}}$$

$$= \frac{R_2SC_1}{(1 + SC_1R_1)(1 + SR_2C_2) + R_2SC_1}$$

$$= \frac{SR_2C_1}{1 + S(R_1C_1 + R_2C_2) + S^2R_1R_2C_1C_2 + SC_1R_2}$$

$$\beta = \frac{j\omega C_1R_2}{1 + j\omega(R_1C_1 + R_2C_2) - \omega^2R_1R_2C_1C_2 + j\omega C_1R_2}$$

$$= \frac{j\omega C_1R_2}{j\omega(R_1C_1 + R_2C_2 + C_1R_2) - (\omega^2R_1R_2C_1C_2 - 1)}$$

Rationalising with expression

Then

$$\beta = \frac{\omega^2 C_1 R_2 (R_1C_1 + R_2C_2 + C_1R_2) + j\omega C_1 R_2 (1 - \omega^2 R_1 R_2 C_1 C_2)}{(1 - \omega^2 R_1 R_2 C_1 C_2) + \omega^2 (R_1 C_1 + R_2 C_2 + R_2 C_1)}$$

$$\Rightarrow \omega (1 - \omega^2 R_1 R_2 C_1 C_2) = 0$$

$$\omega^2 = \frac{1}{R_1 R_2 C_1 C_2}$$

let $R_1 = R_2 = R$

$C_1 = C_2 = C$

$$\omega^2 = \frac{1}{RC}$$

$$2\pi f = \frac{1}{RC}$$

$$f = \frac{1}{2\pi RC}$$

Here $\beta = 1/3$

$|A| \times 1/3 \geq 1$

$|A| \geq 3$

Without any phase shift, gain of amplifier

Advantages :

1. Different frequency ranges can be obtained by varying the 2 capacitor values.
2. Perfect sine wave is possible.
3. It is useful for audio frequency range.

Disadvantages :

1. Poor frequency stability

Quadrature Type Oscillator :

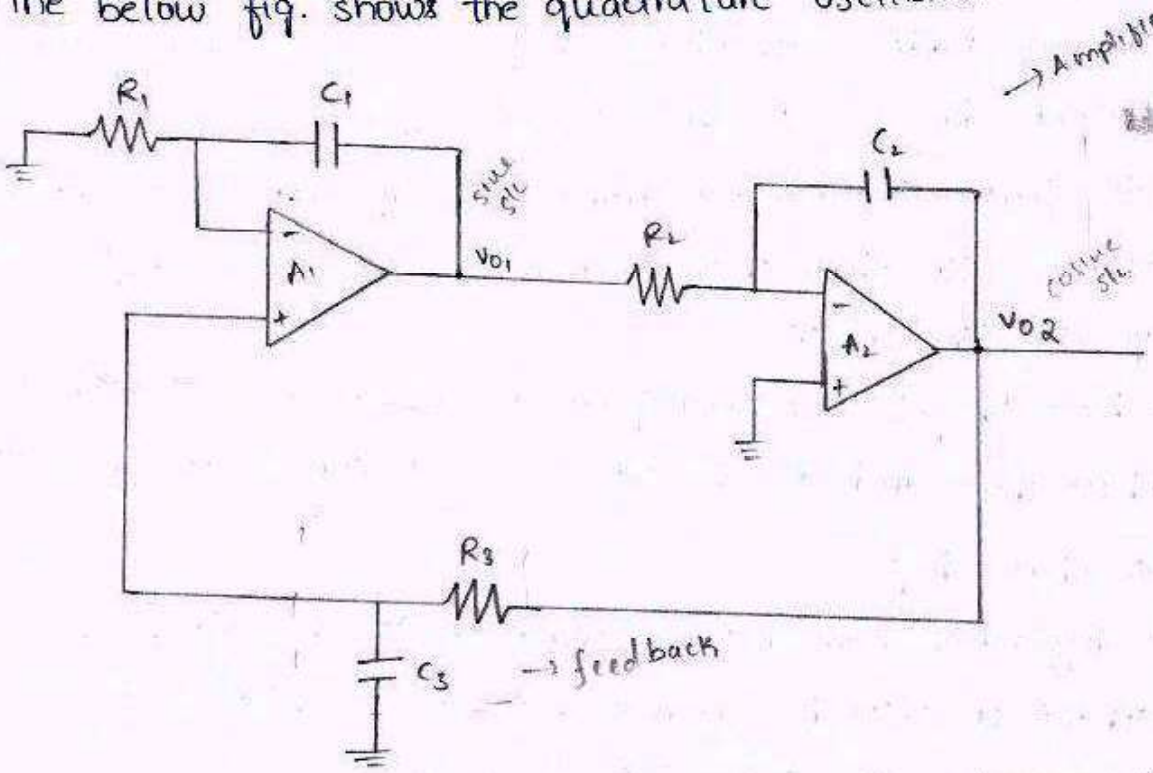
→ If the phase shift b/w the two signals is exactly 90° then it is called as quadrature.

→ But the 2 signals having same frequency.

→ For example, sine & cosine signals are said to be in quadrature.

→ Thus oscillator which generates the 2 signals sine & cosine which are in quadrature is called quadrature oscillator.

→ The below fig. shows the quadrature oscillator.



→ It consists of two op-amps which is called amplifier circuit and one feed back n/w.

→ Let one op-amp is A_1 and other op-amp is A_2.

→ It is required to operate A_1 as an non-inverting integrator while A_2 as pure inverting integrator.

→ Thus R_1C_1 combination is used for op-amp A_1 and R_2C_2 is used for op-amp A_2.

→ The o/p of op-amp A_2 is connected to feed back n/w (R_3C_3)

and voltage across C_3 is given as feedback to the non-inverting i/p of A_1.

→ The op-amp A_1 & A_2 forms amplifier stage of the oscillator while R_3C_3 forms a feedback stage of the oscillator.

Operation :

1. Here op-amp A_2 is an inverting integrator. Due to inverting nature it adds $-180°$ phase shift & due to integrating nature it adds $-90°$ phase shift. So total phase shift introduced due to op-amp A_2 is $-270°$.
2. While feedback n/w R_3C_3 along with op-amp A_1 contributes remaining $-90°$ phase shift and hence total phase shift around a loop is $-360°$ (or) $0°$.
3. We have to get sine waveform at the o/p of A_1 which is $360°$ phase shift and at the o/p of A_2 is cosine waveform which is $-270°$ phase shift.
4. The difference b/w these 2 signals is $90°$ phase shift. Hence this circuit is called Quadrature type oscillator. Due to this it satisfies the Bark Hausen conditions.
5. The frequency f_o produces sine & cosine waveforms is given by

$$f = \frac{1}{2\pi RC}$$

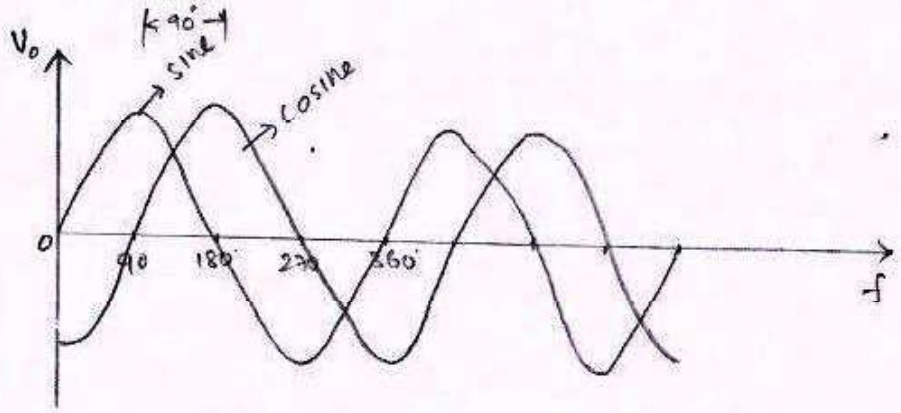
where, $R = R_1 = R_2 = R_3$

$C = C_1 = C_2 = C_3$

Applications :

→ Communication Circuits

→ The corresponding waveforms of quadrature type oscillator is shown in below fig.



Waveform Generators :

→ Generator is a circuit which generates the different types of sinusoidal & non-sinusoidal signals.

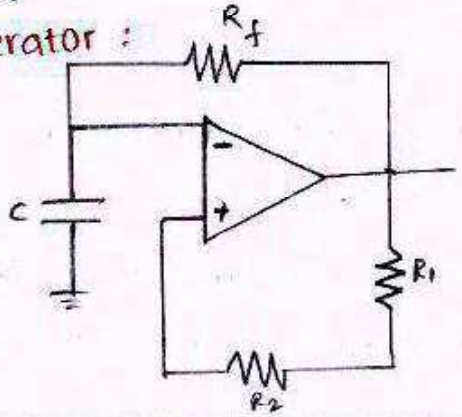
→ If circuit generates the sinusoidal signals then it is called as sinusoidal generator

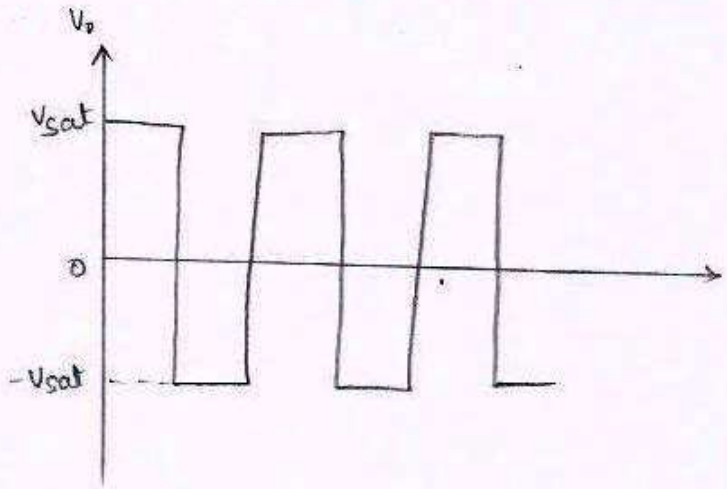
→ If the circuit generates non-sinusoidal signals (square, triangular, sawtooth) then it is called as non-sinusoidal waveform generators.

→ According to definition of non-sinusoidal waveform generators there are different types of generators.

1. Square wave generator
2. Triangular wave generator
3. Sawtooth wave generator

1. Square Wave generator :



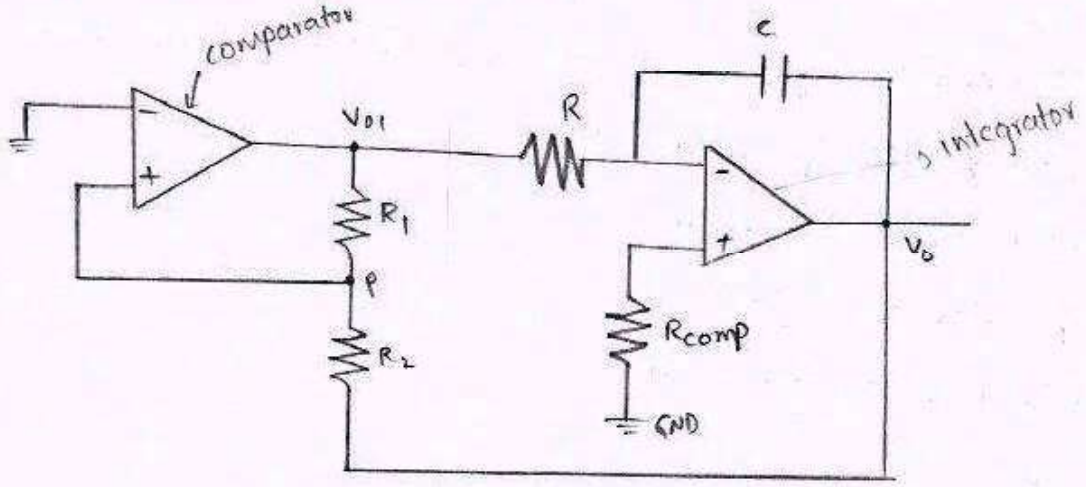


→ To understand the operation of circuit $+V_{sat}$ is available at the o/p. Due to this the capacitor gets charging.

→ The same voltage ($+V_{sat}$) flows through the potential divider n/w due to this some voltage dropped. The dropping voltage appeared at the non-inverting terminal of the comparator.

→ If compare the non-inverting voltage than the inverting voltage which is low. So the comparator can change from $+V_{sat}$ to $-V_{sat}$.

8. Triangular Wave generator :



→ Generally integrator o/p waveform will be triangular if the i/p to it is a square wave. So, the

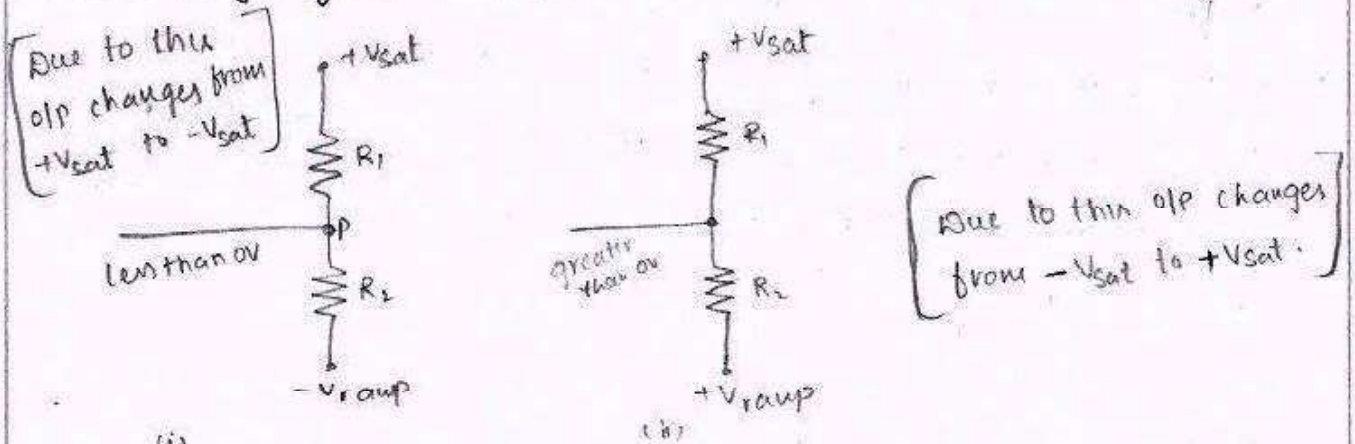
→ So, the triangular wave generator can be implemented by cascading an integrator and a square wave generata. It is shown

in above fig.

→ It consists of two op-amps. First op-amp acts as comparator & second op-amp acts as integrator.

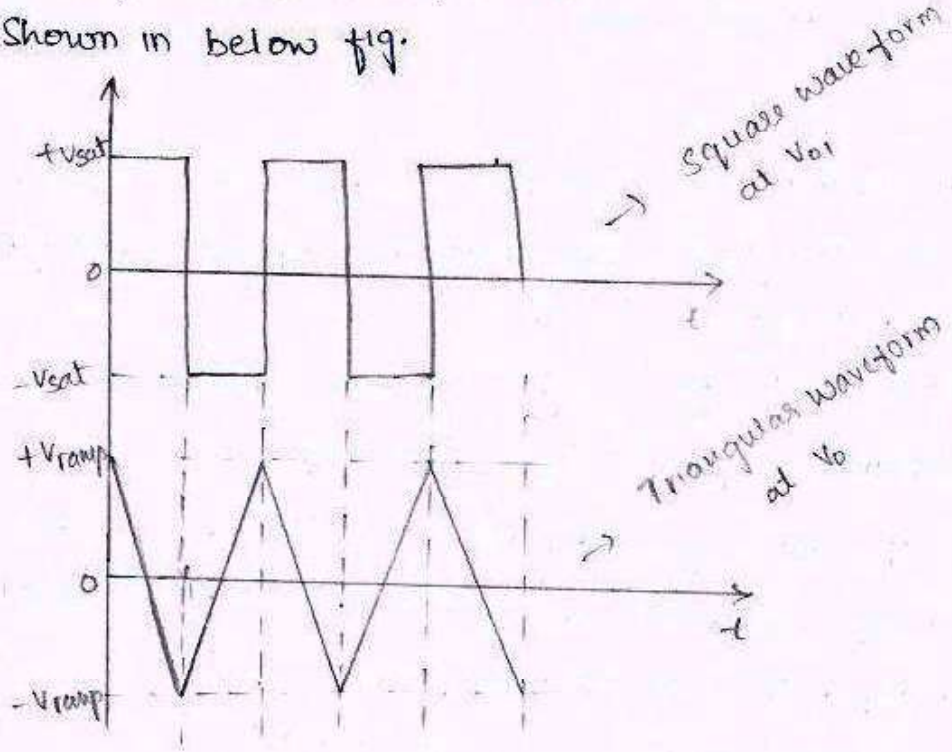
Operation :

1. The comparator compares the voltage at a point 'P' continously with respect to the voltage at the inverting i/p of comparator, which is at ground potential (0v).
2. When the voltage at 'P' goes slightly below 0v, the o/p of comparator will changes from $+V_{sat}$ to $-V_{sat}$.
3. Let us consider the o/p of comparator is $+V_{sat}$, since this voltage is the i/p of integrator then its o/p will be $(-V_{ramp})$ negative going ramp. Thus one end of the potential divider R_1R_2 is at $+V_{sat}$ & other end is at -ve going ramp $(-V_{ramp})$.
4. When the -ve going ramp attains a value say $-V_{ramp}$ the effective voltage at P becomes slightly less than 0v. Then o/p will change from $+V_{sat}$ to $-V_{sat}$.



5. During this time integrating o/p will be +ve going ramp. When the value of +ve going ramp attains $+V_{ramp}$, voltage at 'P' becomes slightly greater than 0V. Due to this o/p of comparator changes from $-V_{sat}$ to $+V_{sat}$. which is shown in fig (ii)

6. This cycle repeates continously and generates triangular wave forms. The corresponding triangular waveform generator shown in below fig.



Sawtooth Wave generator :

→ Sawtooth wave form is a type of linear non-sinusoidal waveform with a triangular shape in which the raise & fall time are different.

→ A ~~fare~~ pure triangular waveform is also linear non-sinusoidal with a triangular shape but it has equal raise & fall time.

→ The sawtooth wave form & triangular waveform graphical representation shown in below fig. Here T_1 is the rise & T_2 is the fall time.



→ $T_1 \neq T_2$

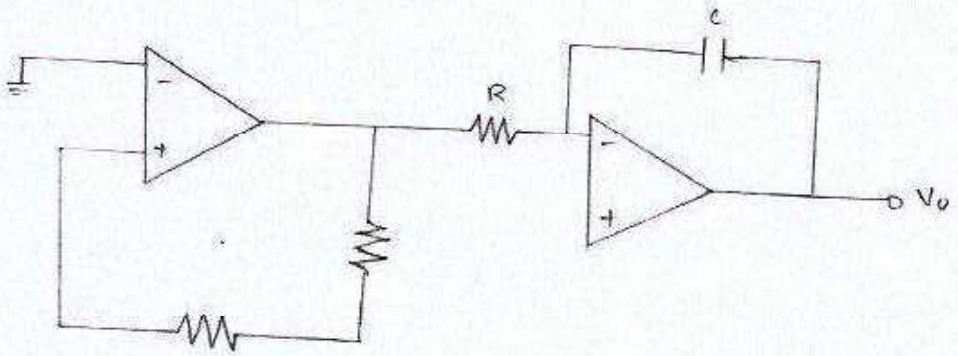
fig: sawtooth wave form



fig: Triangular wave form → $T_1 = T_2$

→ The sawtooth generator can be designed by doing little modifications in the triangular wave generator circuit.

→ In this circuit, the non-inverting terminal of op-amp is grounded. to get sawtooth generate we just need to add a potentiometer & variable resistor at the non-inverting terminal of 2nd op-amp.



→ The circuit consists of comparator, integrator & potentiometer.

→ The potentiometer is used to set the raising time & falling time at the o/p of integrator.

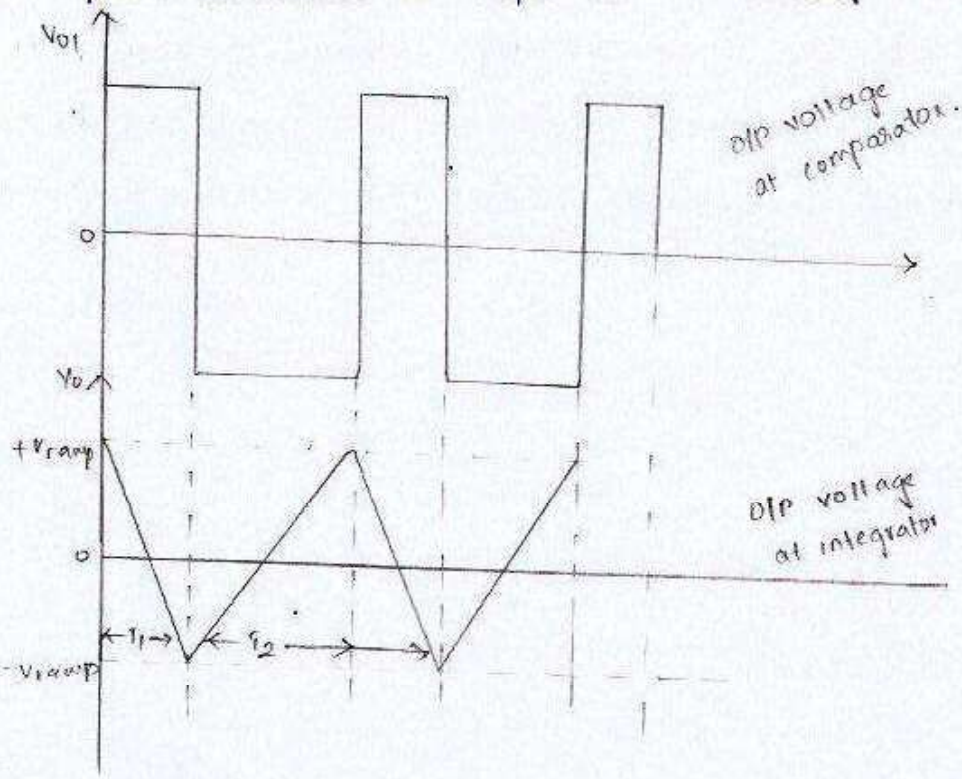
→ If the potentiometer moves towards -ve supply ($-V_{EE}$), the raise time of the saw-tooth becomes larger than the fall time because when potentiometer kept at the -ve supply voltage, the -ve voltage adds at the inverting terminal of op-amp-(2).

→ Due to that, the fall time suddenly downs than the raise time.

→ When the potentiometer moves towards +ve supply ($+V_{CC}$), the fall time becomes larger than the raise time because when potentiometer kept at the +ve supply voltage, the +ve voltage adds at the investing terminal of 2nd op-amp.

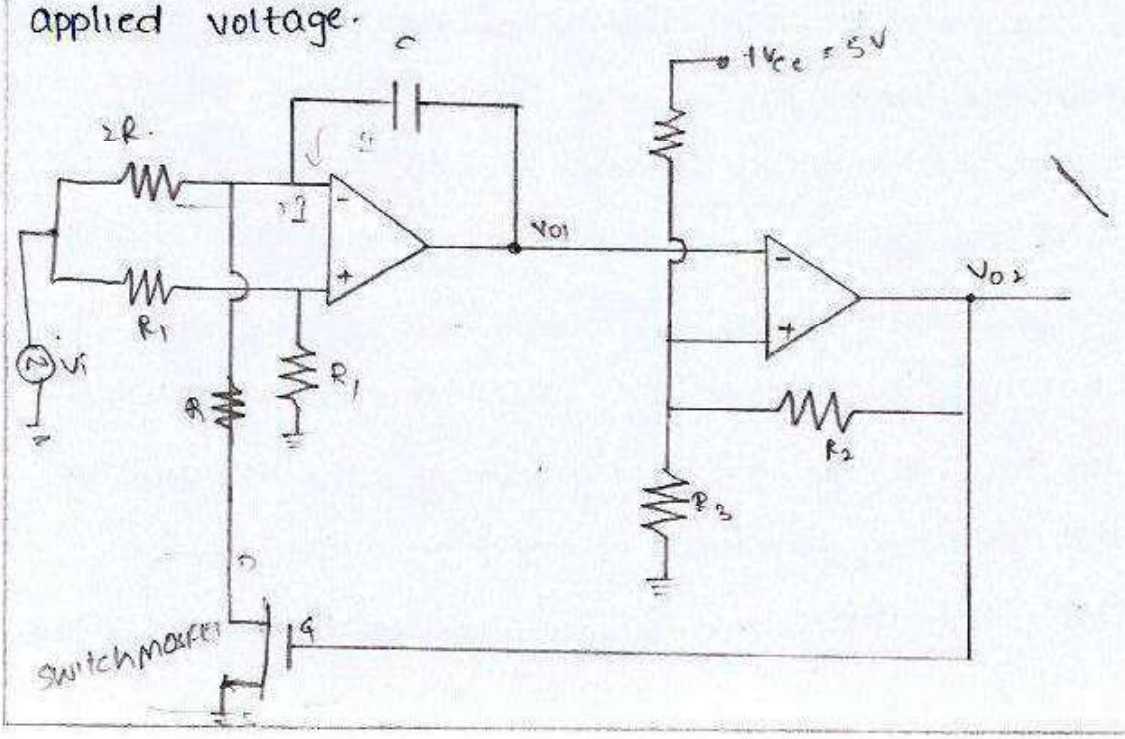
→ Due to that, the raise time suddenly raises than the fall time.

→ The corresponding sawtooth wave generator is shown below when potentiometer is kept at -ve supply voltage.



VCO :

→ VCO is a voltage controlled oscillator. VCO is a circuit in which frequency of oscillations can be controlled by an externally applied voltage.



→ VCO provides the linear relationship blw the applied voltage and the oscillator frequency.
→ Here applied voltage is called as control voltage.
→ Practically VCO is used to produce square & triangular waveforms whose frequency is controlled by control voltage.
→ VCO oscillator circuit is shown in above fig.
→ It consists of two op-amps & one MOSFET switch.
→ First Op-amp can be acts an integrator and other one acts as an comparator.
→ MOSFET can be acts as an switch which is connected blw the o/p of comparator & i/p of integrator.
→ Here comparator o/p is square wave & integrator o/p is triangular wave form.
→ For getting triangular & square waveform, capacitor must charge & discharge. So current I direction through capacitor must keep on alternating.
→ The direction of current I through the capacitor is controlled by the MOSFET switch.
→ But ON & OFF MOSFET switch is controlled by the o/p of comparator. i.e., when o/p of comparator is high, switch is going to ON & o/p of comparator is low. switch is going to OFF.
→ To understand the operation let us assume the o/p of comparator is low. Therefore switch is off. Hence entire current I has to pass through the capacitor 'C' & the voltage V_{01} starts to decreasing due to polarity of voltage across the capacitor.

Hence V_{o1} ramps down.

→ When V_{o1} becomes zero, then comparator changes its state & o/p V_{o2} becomes high.

→ Due to this, switch becomes ON & the resistance R gets connected to the ground. So now the total current 2I i.e., one current I supplied by input & remaining current I must supplied by capacitor flows through the ground.

→ Thus the direction of current through the capacitor gets reversed. Therefore V_o, voltage starts ramping up i.e., starts increasing.

→ It may be noted that capacitor current I gets maintained at same magnitude only its direction reversed. Therefore triangular waveform available at V_{o1}.

→ As soon as V_{o1} reaches the level of V_{o2}, the comparator changes its state making switch is off.

→ Thus the current I through the capacitor again reverse its direction & V_{o1} ramps down. Thus the cycle continous.

→ Corresponding waveforms shown in below fig.

